ZEERBOX · RISK REGISTER AND HAZARD ANALYSIS

ZeerBox risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade) with project risks, preliminary hazard analysis and a first-pass design FMEA of option A

ZeerBox risk register and hazard analysis

The largest project risk is that the store is designed before anyone who would use it has been asked: size, ownership, crops and season are all proposals. The largest hazards are people inside a closed room (entrapment and air quality, the latter worst when the battery is flat, ZBX-DEC-001 D3), warm recirculating water (*Legionella*), and a roof on posts in a strong wind; the highest design FMEA risks are the pad drying or scaling so that the store stops cooling without the user noticing, and a flat battery leaving the room unventilated. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be found) to 5 (unlikely to be found).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	Requirements do not match users' needs	No co-design session yet; targets set from literature (ZBX-URS-001)	Wrong size, cost or season; a store that sits unused, as some solar cold rooms did (ZBX-PRB-001)	4	5	Project lead	Approach the first candidate partner (ZBX-DEC-001) before any TRL 4 work; revise URS and PRS from sessions
P2	Fan and pad curves differ from the assumed ones	Airflow and effectiveness rest on assumed curves (STM S3, S5)	Less airflow or effectiveness, warmer and drier store	3	3	Design lead	Get supplier charts (ZBX-DEC-001 to confirm 2); rerun CAL; a 185 mm pad or a third fan are the levers
P3	Humidity below target in the unfavorable case	The air warms 2 to 3 K across the store and loses RH (PR-003 at risk)	Leafy produce wilts	3	3	Design lead	Covered or lined crates (decided); measure at TRL 4
P4	No dry, pest-proof rice husk near the site	Local supply unknown	Fall back to dry sand, about 0.3 K warmer and about 1.8 t heavier	3	2	Partner	Confirm supply (to confirm 5)
P5	Cellulose pads not sold in the region	Pads come from poultry or greenhouse suppliers, perhaps only in a capital city	Delay or higher cost; replacements every few years	3	3	Procurement	Survey suppliers with the partner (PR-010); price in MKT
P6	Kit cost rises past the value-engineering target	Prices are indicative, not quoted; kit already at USD 300	Over target	3	2	Project lead	Quotes at TRL 4; savings listed in ZBX-DEC-001 (steel angle frames, about USD 8)
P7	The store is used in the rainy season and disappoints	Evaporative cooling gives 2 to 3 K at 75 % RH	Users lose trust	3	3	Partner	Lamp shows the mode; set expectations in training; ask whether rainy-season use matters (ZBX-PRB-001)
P8	Recommended concept is not chosen	Options within 10 % (ZBX-OPT-001);	Redesign to option B or C	2	4	Amish	Decision requested in ZBX-DDR-004

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
		concept choice is Amish's					

Preliminary hazard analysis

Table 2. Hazards (option A).

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Entrapment in the store	Door closed on a person inside	Heat stress, fear, injury	4	2	Inside release latch, no outside lock (PR-009); door opens outward	Low
H2	Poor air in a closed store	Produce respiration; fans off when the battery is flat for more than about 40 h, or hold-mode flow stalled by wind on the pad wall (independent review); door shut	About 2.3 % CO_2 in 12 h sealed (STM S36), dizziness	3	2	Hold mode keeps about 8 air changes an hour while powered (STM S28); safety note to air the store before entering; passive vent proposed (ZBX-DEC-001 D3)	Medium until D3 is decided
H3	*Legionella* and other bacteria in the sump and pad	Warm recirculating water, sediment, biofilm; droplets from the pad	Respiratory infection	4	2	Covered, shaded sump; daily bleed and daily pad dry-out; fans run 10 min before entry (D5); weekly drain and clean; weekly shock chlorination proposed (D8); water never used for drinking or washing produce (WHO 2007, ZBX-REG-001 G9)	Medium, depends on routine
H4	Mold on produce and walls	Damp store, rotten produce left in	Spoiled produce; spores	2	3	Guard mode stops the pump at 95 % RH for 60 min (PR-005); weekly cleaning	Low
H5	Roof lifted by wind	Gust above the 30 m/s design case (the footing factor falls below 1.5 above about 32 m/s); edge-zone pressures; ties not rated	Falling sheets and timber	5	2	Footings 1.8 times the uplift; ties rated 2 kN; sheets screwed to every purlin; confirm the local design wind speed before building (to confirm 6, ZBX-DEC-001 D7)	Medium until the wind speed is confirmed
H6	Masonry collapse	Poor workmanship; cracks; soft soil	Crushing	5	1	Strip footing, competent mason, seasonal	Low

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
						inspection; bearing 16 kPa on firm soil	
H7	Battery fire or burn	Short circuit, charging out of the 0 to 45 °C range, damaged cell	Fire, burns	4	1	LiFePO4 with BMS; DC-rated 15 A battery fuse on 2.5 mm² leads and 5 A load fuse (ZBX-DEC-001 D1, D5); shaded, ventilated box; location inside the store proposed (D6)	Low
H8	Cable overheating	Load-circuit fault protected only by the 15 A battery fuse (found by STM S16)	Fire in the roof space	4	2	Fixed: 5 A load fuse within the 13.8 A capacity of the 1.5 mm² cable (D1)	Low
H9	Fan blade contact	Hand in the fan during cleaning	Cuts	2	2	Guards on both faces; isolate power before cleaning	Low
H10	Rice husk fire	Hot work near the walls; damaged conduit	Smouldering fire in the cavity	4	1	Cavity capped with mortar; no hot work near the walls; solar wiring in conduit at the wall top	Low
H11	Food safety misunderstanding	Users treat the store as a cold room	Illness from meat, fish or milk stored warm	4	2	Clear labelling and training: not for meat, fish, milk or medicines (ZBX-PRC-001); Codex practice (REG G11)	Medium

Design FMEA (first pass, P3)

Table 3. Failure modes of option A. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Cellulose pad (line 6)	Wet the incoming air	Dry patches or scale on the pad	Warmer, drier store; user may not notice	Pump flow too low, blocked header holes, hard water	4	3	4	48	Bleed sized for 4 cycles of concentration (D5); log supply-air RH and alarm on a high supply temperature at TRL 4; clean header holes in weekly routine
F2	Pump and float switch (line 7)	Lift water to the header	Pump runs dry or stops	Pad dries; pump burns out	Sump not refilled; float stuck	3	3	3	27	Float switch interlock (guard mode); sump level visible; weekly check
F3	Exhaust fans (line 8)	Move about 560 m³/h	Fan stops or slows	No cooling; rising CO_2	Dust in bearings; connector corrosion	4	2	3	24	Controller checks fan current; spare fan; guards cleaned
F4	Inside and outside sensors (line 9)	Pick the mode	Sensor drift or failure	Wrong mode (wet pad in humid weather or no cooling)	Dust, condensation, radiation on the outside sensor	3	3	3	27	Six-plate shield; plausibility check between sensors; replace yearly
F5	Battery and charge controller (line 11)	Store energy for the evening and cloud	Battery flat or BMS cut-off	Fans stop; sealed room (H2)	Long cloud, harmattan haze, ageing, BMS stopping charge above 45 °C	4	3	3	36	Low-battery guard mode; passive vent proposed (D3); battery inside the store proposed (D6)
F6	Load fuse (line 11)	Protect the load cable	Wrong fuse fitted (15 A in place of 5 A)	Cable overheats on a fault	Generic fuses look alike	4	2	4	32	Label the holder "5 A"; fuse rating in the build plan wiring step
F7	Door seal and latch (line 5)	Keep warm air out; open from inside	Seal worn; latch jams	More heat load; entrapment risk	Wear, dust, warping	4	2	2	16	Inside release with no lock; seasonal check
F8	Rice husk fill (line 2)	Insulate the walls	Fill gets wet or eaten	Higher wall gain; pests	Leaking capping course; rodents	2	2	4	16	Damp-proof course, lime, mortar cap, mesh at openings
F9	Hurricane ties and roof screws (lines 4, 17)	Hold the roof down	Tie or screw pulls out	Roof damage in a storm	Undersized or missing ties; rotten timber	5	2	3	30	Ties rated 2 kN (to confirm 4); count every tie at handover

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F10	Gutter and return hose (line 6)	Return pad water to the sump	Gutter overflows or hose blocks	Water on the floor and wall; sump empties	Debris, algae	2	3	2	12	Weekly clean; mesh at the outlet

The pad drying or scaling unnoticed (F1, RPN 48), a flat battery leaving the room unventilated (F5, RPN 36) and a wrong fuse (F6, RPN 32) are the top design risks; they are addressed by the bleed (D5), the decisions proposed in D3 and D6, the build plan and logging at TRL 4.